Cetacean Communication Station

The Making of the Cetacean Interface

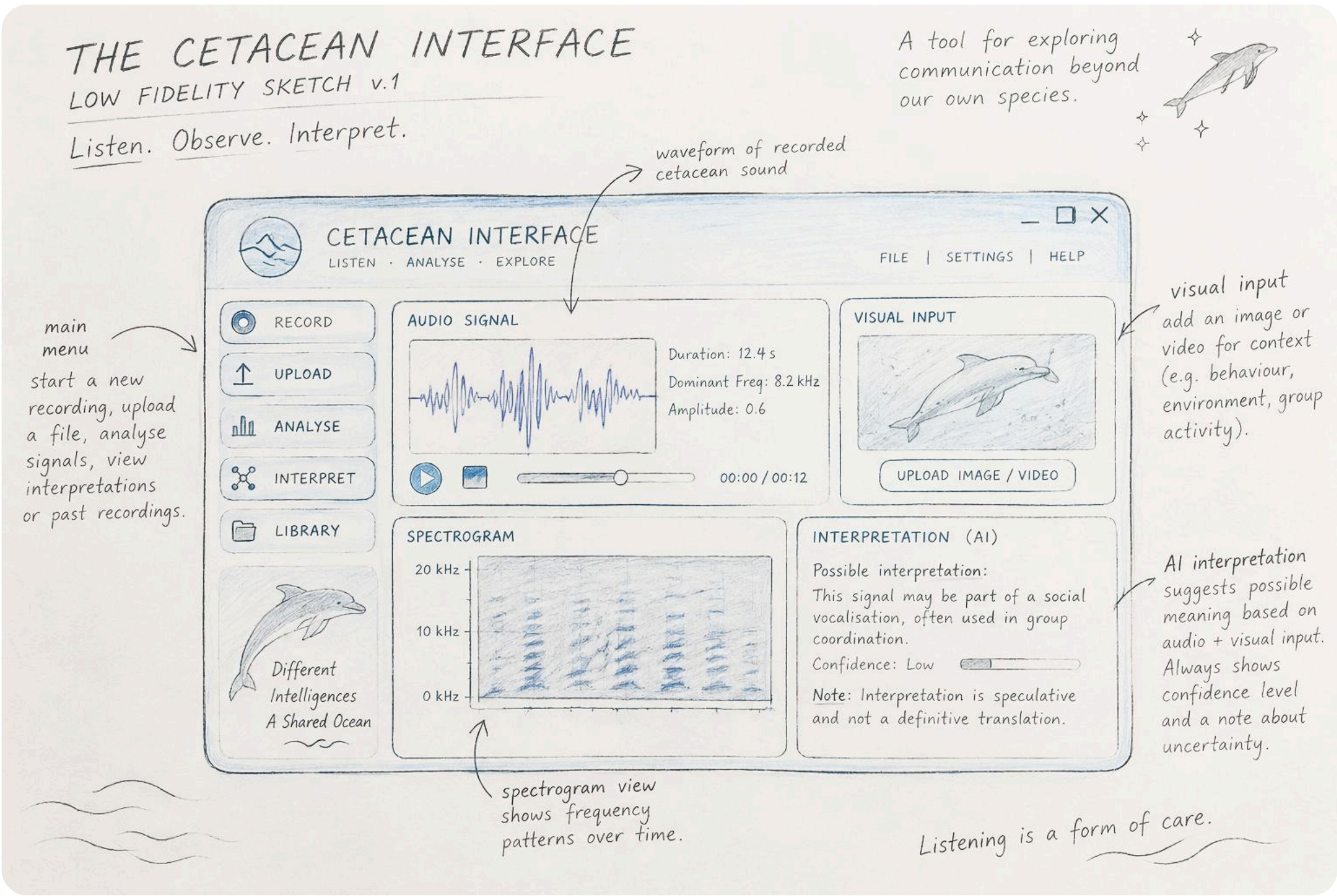
by Ashani Gorakhnath

Different Intelligences
Shared Planet
SAME OCEAN DIFFERENT
Marine Biology
Acoustic Communication
More-Than-Human Futures
Ocean Rights
Design Fiction
CETACEAN INTERFACE
LISTENING...
SIGNAL DETECTED
Analysing acoustic patterns...
CURIOSITY BUILDS BRIDGES
HUMAN A KINDER
Same Ocean Different Story

Concept note

Cetacean Communication Station

The Cetacean Interface is an experimental digital station designed to help Keto observe and interpret cetacean signals. It combines audio input, visual observation, acoustic analysis, and AI generated interpretations within a single interface.
Inspired by Y2K Frutiger Aero aesthetics, its visual language evokes a once optimistic relationship between nature and technology. The interface transforms complex signals into visual and spoken outputs, while making the uncertainty of understanding another intelligence part of the experience.


THE CETACEAN INTERFACE
LOW FIDELITY SKETCH v.1
Listen. Observe. Interpret.
A tool for exploring communication beyond our own species.
waveform of recorded cetacean sound
CETACEAN INTERFACE
LISTEN · ANALYSE · EXPLORE
FILE | SETTINGS | HELP
main menu
start a new recording, upload a file, analyse signals, view interpretations or past recordings.
RECORD
UPLOAD
ANALYSE
INTERPRET
LIBRARY
Different Intelligences A Shared Ocean
AUDIO SIGNAL
Duration: 12.4 s
Dominant Freq: 8.2 kHz
Amplitude: 0.6
00:00 / 00:12
VISUAL INPUT
UPLOAD IMAGE / VIDEO
visual input add an image or video for context (e.g. behaviour, environment, group activity).
SPECTROGRAM
20 kHz
10 kHz
0 kHz
INTERPRETATION (AI)
Possible interpretation:
This signal may be part of a social vocalisation, often used in group coordination.
Confidence: Low
Note: Interpretation is speculative and not a definitive translation.
AI interpretation suggests possible meaning based on audio + visual input. Always shows confidence level and a note about uncertainty.
spectrogram view shows frequency patterns over time.
Listening is a form of care.

Initial idea

Iteration 1: Imagining the communication station

Concept visualisation · Physical and digital artefact

The initial idea was to imagine how Keto might create a tool to communicate with cetaceans in a future where humans are separated from the ocean. I began exploring the Cetacean Interface as a physical communication station, combining a digital screen with my artefact.
The visualisation helped establish the artefact's material language: a bright, aquatic colour palette, translucent surfaces, and a playful, retro-futuristic aesthetic inspired by Frutiger Aero. The device was imagined as something Keto could assemble using available technology, drawing from an earlier generation of technological optimism.
At this stage, the focus was on imagining the form and atmosphere of the artefact rather than its actual functionality.

Key decision:

To treat the interface as a bridge between human technology and cetacean intelligence, while retaining the handmade, experimental quality of Keto's invention.

Reflection:

The concept established the visual identity, but I needed to explore how the artefact could function as an interactive system rather than remain a speculative object.

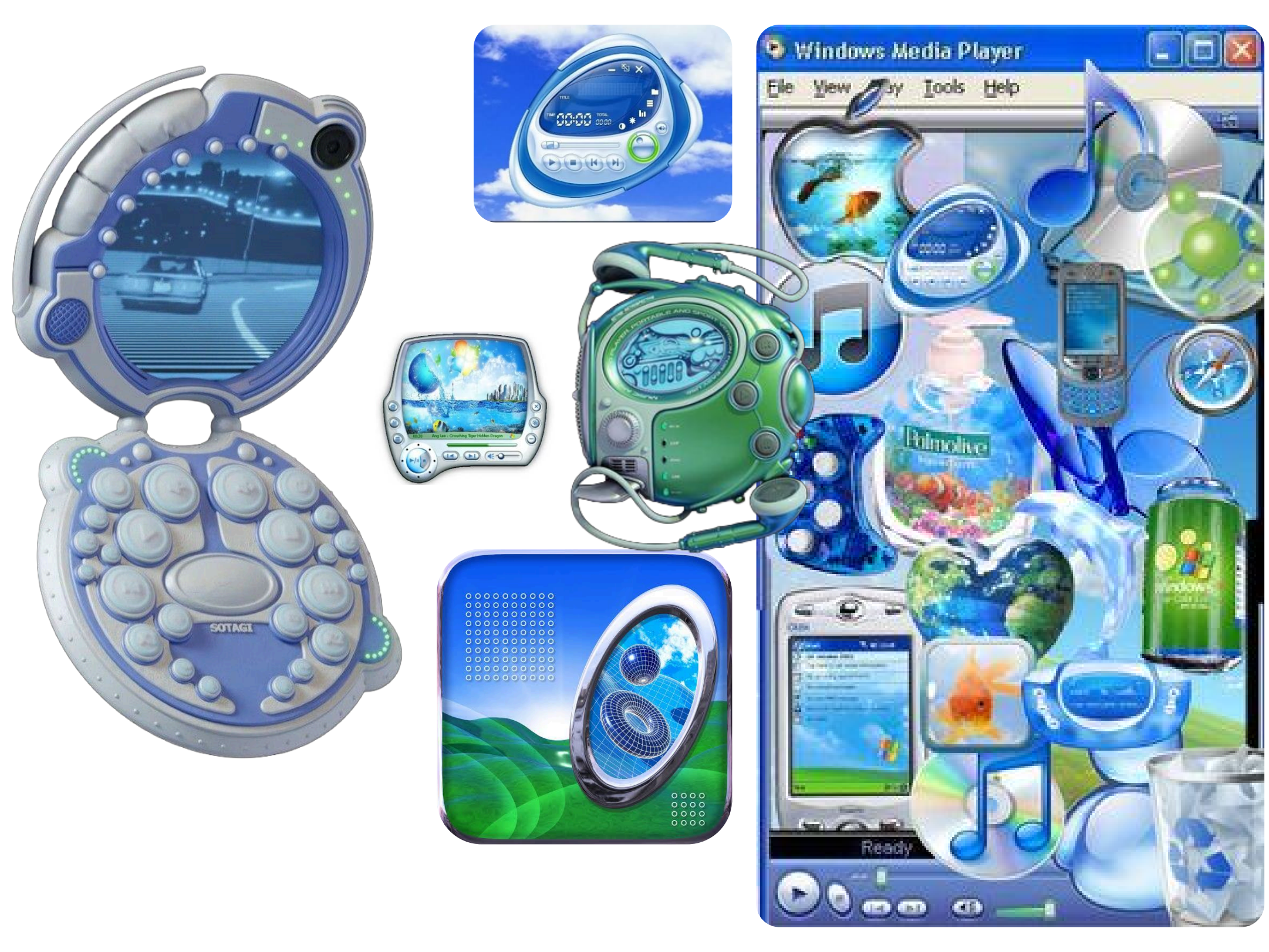
Building blocks

Visual reference

A Future Remembered Through Y2K Technology

The visual language of the Cetacean Interface draws from Frutiger Aero, transparent technology, and early digital media, particularly the glossy skins and playful interfaces of MP3 players and media software.
These references evoke a time when technology was imagined as fluid, optimistic, and closely connected to nature. I reinterpreted this aesthetic through aquatic blues, translucent panels, soft gradients, rounded forms, and luminous interface elements.
The contrast between this nostalgic visual language and the uncertain future of human–cetacean communication reflects Keto's attempt to use the technological optimism of the past to navigate an unfamiliar world.

Forms: Rounded panels, soft edges, bubble-like shapes
Materials: Glass-like transparency, glossy surfaces, layered translucency
Interaction: Compact media controls, audio visualisation, status indicators
Atmosphere: Nostalgic, experimental, optimistic, slightly retro-futuristic


Windows Media Player
File
View
Tools
Help
Ready

Translating the concept into a digital interface

Interation 2: Streamlit interface

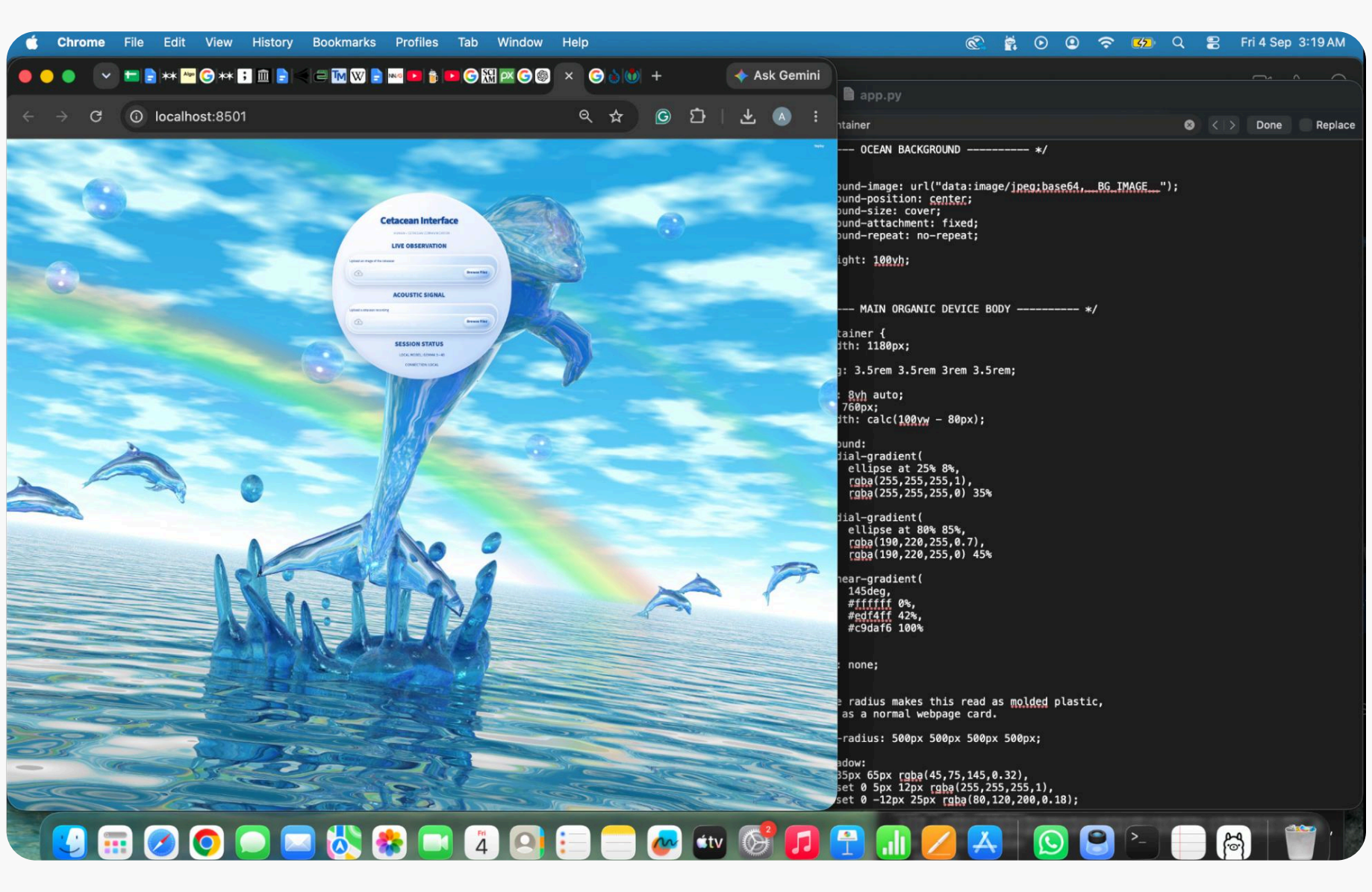
The next iteration translated the imagined communication station into a functional digital prototype. Instead of focusing on the physical device, I began developing an interface that could bring together audio input, visual observation, acoustic analysis, and AI-generated interpretation.
The interface was organised into distinct sections for live observation, acoustic signal input, a spectrogram, and signal analysis. I introduced controls for live multimodal capture, recording uploads, camera selection, and analysing the captured signal.
This iteration shifted the artefact from a visual concept towards an interactive experience. The user could engage with the system, provide inputs, and observe how the interface processed and presented information.

Reflection:

The concept established the visual identity, but I needed to explore how the artefact could function as an interactive system rather than remain a speculative object.

Key decision:

- Organise the interface around the sequence of observing, listening, and analysing.
- Use a spectrogram and acoustic measurements to make sound visible.
- Retain the Frutiger Aero visual language through glossy surfaces, soft gradients, and aquatic colours.
- Include system status indicators to communicate whether the interface is ready, listening, or processing.


Chrome
File
Edit
View
History
Bookmarks
Profiles
Tab
Window
Help
Fri 4 Sep 3:19 AM
Ask Gemini
localhost:8501
app.py
Done
Replace
Cetacean Interface
LIVE OBSERVATION
ACOUSTIC SIGNAL
SESSION STATUS

```
-- OCEAN BACKGROUND --------- */
ound-image: url("data:image/jpeg;base64,__BG_IMAGE__");
ound-position: center;
ound-size: cover;
ound-attachment: fixed;
ound-repeat: no-repeat;
ight: 100vh;

-- MAIN ORGANIC DEVICE BODY --------- */
ainer {
dth: 1180px;
: 3.5rem 3.5rem 3rem 3.5rem;
 8vh auto;
760px;
dth: calc(100vw - 80px);
und:
ial-gradient(
 ellipse at 25% 8%,
 rgba(255,255,255,1),
 rgba(255,255,255,0) 35%
ial-gradient(
 ellipse at 80% 85%,
 rgba(190,220,255,0.7),
 rgba(190,220,255,0) 45%
near-gradient(
 145deg,
 #ffffff 0%,
 #edf4ff 42%,
 #c9daf6 100%
 none;
 radius makes this read as molded plastic,
 as a normal webpage card.
-radius: 500px 500px 500px 500px;
adow:
35px 65px rgba(45,75,145,0.32),
set 0 5px 12px rgba(255,255,255,1),
set 0 -12px 25px rgba(80,120,200,0.18);
```

Refining the interaction and making uncertainty visible

Iteration 3: Interaction and system feedback

Concept visualisation · Physical and digital artefact

As the interface developed, the focus shifted towards making the interaction clearer and more responsive. I refined the relationship between capturing a signal and analysing it, so that analysis would happen only when the user chose to initiate it.

The prototype also developed clearer status feedback, including indicators for sensor readiness, connection state, and signal availability. These elements helped communicate what the system was doing at different stages of the interaction.

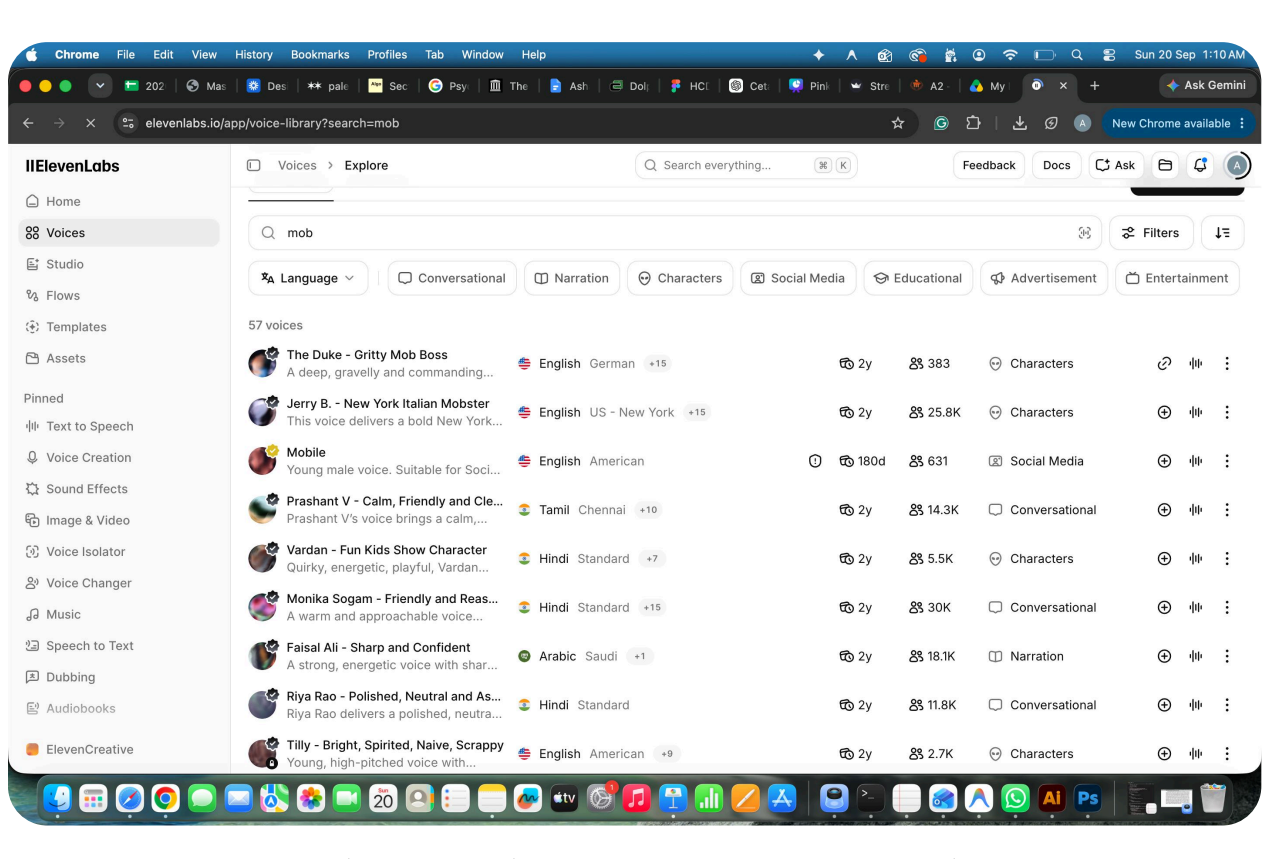
A central consideration was how to present AI-generated interpretations without suggesting that the system had achieved a reliable translation of cetacean communication.

Key decision:

o frame the output as an experimental interpretation rather than a definitive translation. For narrative comedic purposes I decided that the converted audio would utilise 1920s mobspeak.

Reflection:

The limitations of the technology became part of the artefact's meaning. The interface could make signals visible and generate possible interpretations, but the gap between signal and meaning remained unresolved.

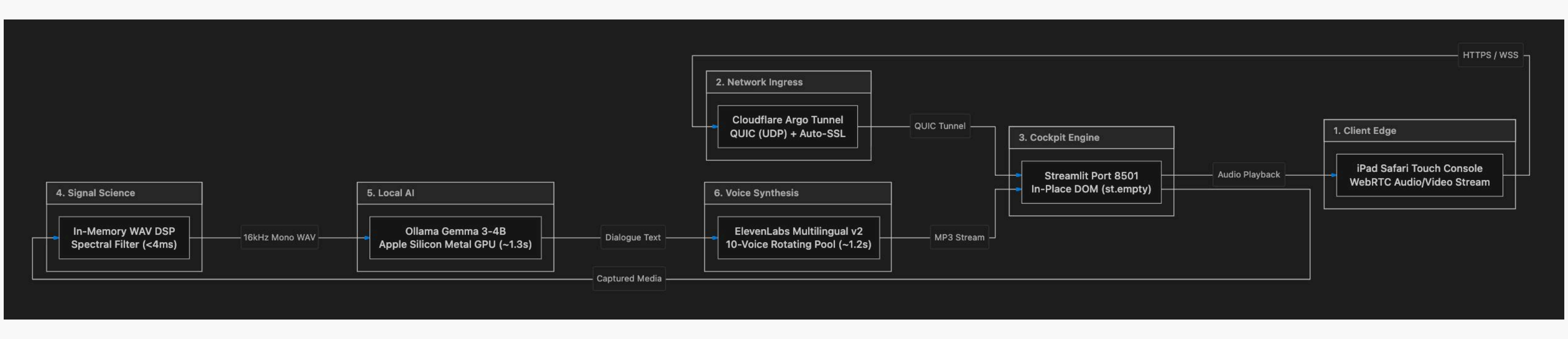

elevenlabs.io/app/voice-library?search=mob
IIElevenLabs
Home
Voices
Studio
Flows
Templates
Assets
Voices > Explore
mob
Language
Conversational
Narration
Characters
Social Media
Educational
Advertisement
Entertainment
57 voices

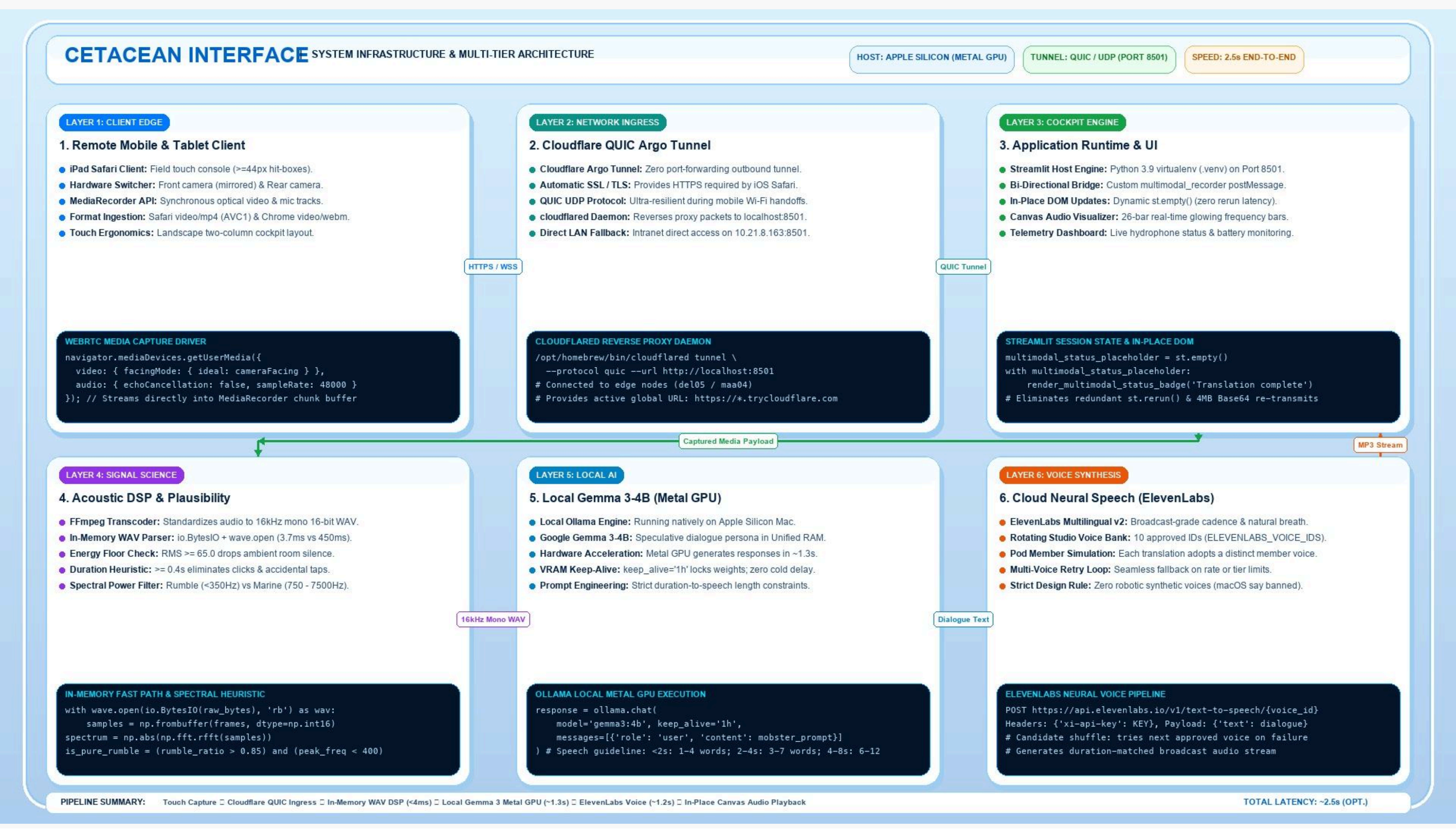
The Duke - Gritty Mob Boss A deep, gravelly and commanding...	English German +15	2y	383	Characters
Jerry B. - New York Italian Mobster This voice delivers a bold New York...	English US - New York +15	2y	25.8K	Characters
Mobile Young male voice. Suitable for Soci...	English American	180d	631	Social Media
Prashant V - Calm, Friendly and Cle... Prashant V's voice brings a calm,...	Tamil Chennai +19	2y	14.3K	Conversational
Vardan - Fun Kids Show Character Quirky, energetic, playful, Vardan...	Hindi Standard +7	2y	5.5K	Characters
Monika Sogam - Friendly and Reas... A warm and approachable voice...	Hindi Standard +15	2y	30K	Conversational
Faisal Ali - Sharp and Confident A strong, energetic voice with shar...	Arabic Saudi +1	2y	18.1K	Narration
Riya Rao - Polished, Neutral and As... Riya Rao delivers a polished, neutra...	Hindi Standard	2y	11.8K	Conversational
Tilly - Bright, Spirited, Naive, Scrappy Young, high-pitched voice with...	English American +8	2y	2.7K	Characters

Setting up an API via Eleven Labs to generate mobster audio

Layer-by-Layer Technical Breakdown

Layer	Technology Stack	Core Function & Infrastructure Duties
Client / Edge Layeri	Pad Safari / Desktop Web (WebRTC, MediaRecorder API, HTML5 Canvas)	Ingests camera (front/rear toggling) and microphone streams; handles touch ergonomics (≥44px≥44px) and renders the 2-column Frutiger Aero cockpit dashboard.
Network & Ingress Layer	Cloudflare Argo Tunnels (cloudflared, QUIC over UDP, HTTP/2 fallback)	Provides zero-port-forwarding outbound reverse proxying; automatically issues trusted SSL/TLS certificates strictly required by iOS Safari for microphone/camera access.
Application Runtime	Streamlit (Python 3.9, .venv, port 8501, postMessage component bridges)	Hosts the application server, coordinates state sessions, in-place st.empty() DOM updates, and custom Web Audio API Canvas visualizers.
Acoustic DSP & Plausibility	FFmpeg + NumPy FFT (np.fft.rfft, in-memory wave.open, @st.cache_data)	Transcodes video to 16kHz mono WAV; in-memory parser (3.7ms3.7ms); filters low-frequency rumble (<350Hz<350Hz) vs marine whistle bands (750Hz–7500Hz750Hz–7500Hz).
Local Generative AI	Ollama + Google Gemma 3-4B (Apple Silicon Metal GPU, keep_alive="1h")	Performs on-device, private speculative dialogue generation in ~1.3s with pre-warmed weights in Unified Memory; enforces strict duration-to-speech length limits.
Cloud Neural Audio	ElevenLabs Multilingual v2 (ELEVENLABS_VOICE_IDS, multi-voice retry loop)	Synthesizes broadcast-grade speech using a rotating pool of 10 studio voices to simulate distinct cetacean pod members, with zero robotic fallback.


HTTPS / WSS
2. Network Ingress
Cloudflare Argo Tunnel QUIC (UDP) + Auto-SSL
QUIC Tunnel
3. Cockpit Engine
Streamlit Port 8501 In-Place DOM (st.empty)
Audio Playback
1. Client Edge
iPad Safari Touch Console WebRTC Audio/Video Stream
4. Signal Science
In-Memory WAV DSP Spectral Filter (<4ms)
16kHz Mono WAV
5. Local AI
Ollama Gemma 3-4B Apple Silicon Metal GPU (~1.3s)
Dialogue Text
6. Voice Synthesis
ElevenLabs Multilingual v2 10-Voice Rotating Pool (~1.2s)
MP3 Stream
Captured Media


HOST: APPLE SILICON (METAL GPU)
TUNNEL: QUIC / UDP (PORT 8501)
SPEED: 2.5s END-TO-END
LAYER 1: CLIENT EDGE
LAYER 2: NETWORK INGRESS
LAYER 3: COCKPIT ENGINE
HTTPS / WSS
QUIC Tunnel
Captured Media Payload
MP3 Stream
LAYER 4: SIGNAL SCIENCE
LAYER 5: LOCAL AI
LAYER 6: VOICE SYNTHESIS
16kHz Mono WAV
Dialogue Text

CETACEAN INTERFACE SYSTEM INFRASTRUCTURE & MULTI-TIER ARCHITECTURE

1. Remote Mobile & Tablet Client

- **iPad Safari Client:** Field touch console (>=44px hit-boxes).
- **Hardware Switcher:** Front camera (mirrored) & Rear camera.
- **MediaRecorder API:** Synchronous optical video & mic tracks.
- **Format Ingestion:** Safari video/mp4 (AVC1) & Chrome video/webm.
- **Touch Ergonomics:** Landscape two-column cockpit layout.

WEBRTC MEDIA CAPTURE DRIVER

```
navigator.mediaDevices.getUserMedia({
  video: { facingMode: { ideal: cameraFacing } },
  audio: { echoCancellation: false, sampleRate: 48000 }
}); // Streams directly into MediaRecorder chunk buffer
```

2. Cloudflare QUIC Argo Tunnel

- **Cloudflare Argo Tunnel:** Zero port-forwarding outbound tunnel.
- **Automatic SSL / TLS:** Provides HTTPS required by iOS Safari.
- **QUIC UDP Protocol:** Ultra-resilient during mobile Wi-Fi handoffs.
- **cloudflared Daemon:** Reverses proxy packets to localhost:8501.
- **Direct LAN Fallback:** Intranet direct access on 10.21.8.163:8501.

CLOUDFLARED REVERSE PROXY DAEMON

```
/opt/homebrew/bin/cloudflared tunnel \
  --protocol quic --url http://localhost:8501
# Connected to edge nodes (del05 / maa04)
# Provides active global URL: https://*.trycloudflare.com
```

3. Application Runtime & UI

- **Streamlit Host Engine:** Python 3.9 virtualenv (.venv) on Port 8501.
- **Bi-Directional Bridge:** Custom multimodal_recorder postMessage.
- **In-Place DOM Updates:** Dynamic st.empty() (zero rerun latency).
- **Canvas Audio Visualizer:** 26-bar real-time glowing frequency bars.
- **Telemetry Dashboard:** Live hydrophone status & battery monitoring.

STREAMLIT SESSION STATE & IN-PLACE DOM

```
multimodal_status_placeholder = st.empty()
with multimodal_status_placeholder:
    render_multimodal_status_badge('Translation complete')
# Eliminates redundant st.rerun() & 4MB Base64 re-transmits
```

4. Acoustic DSP & Plausibility

- **FFmpeg Transcoder:** Standardizes audio to 16kHz mono 16-bit WAV.
- **In-Memory WAV Parser:** io.BytesIO + wave.open (3.7ms vs 450ms).
- **Energy Floor Check:** RMS >= 65.0 drops ambient room silence.
- **Duration Heuristic:** >= 0.4s eliminates clicks & accidental taps.
- **Spectral Power Filter:** Rumble (<350Hz) vs Marine (750 - 7500Hz).

IN-MEMORY FAST PATH & SPECTRAL HEURISTIC

```
with wave.open(io.BytesIO(raw_bytes), 'rb') as wav:
    samples = np.frombuffer(frames, dtype=np.int16)
spectrum = np.abs(np.fft.rfft(samples))
is_pure_rumble = (rumble_ratio > 0.85) and (peak_freq < 400)
```

5. Local Gemma 3-4B (Metal GPU)

- **Local Ollama Engine:** Running natively on Apple Silicon Mac.
- **Google Gemma 3-4B:** Speculative dialogue persona in Unified RAM.
- **Hardware Acceleration:** Metal GPU generates responses in ~1.3s.
- **VRAM Keep-Alive:** keep_alive='1h' locks weights; zero cold delay.
- **Prompt Engineering:** Strict duration-to-speech length constraints.

OLLAMA LOCAL METAL GPU EXECUTION

```
response = ollama.chat(
    model='gemma3:4b', keep_alive='1h',
    messages=[{'role': 'user', 'content': mobster_prompt}]
) # Speech guideline: <2s: 1-4 words; 2-4s: 3-7 words; 4-8s: 6-12
```

6. Cloud Neural Speech (ElevenLabs)

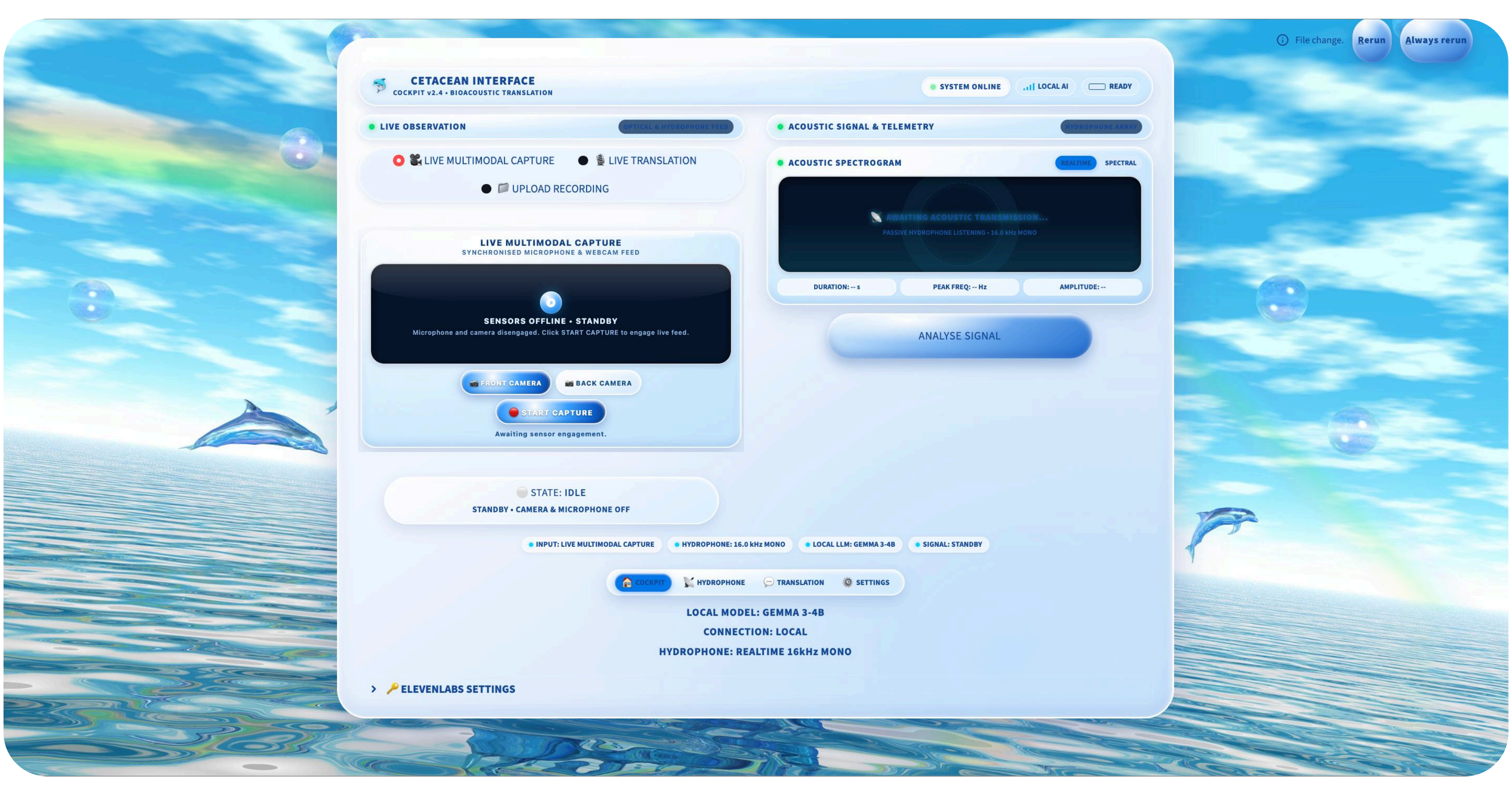
- **ElevenLabs Multilingual v2:** Broadcast-grade cadence & natural breath.
- **Rotating Studio Voice Bank:** 10 approved IDs (ELEVENLABS_VOICE_IDS).
- **Pod Member Simulation:** Each translation adopts a distinct member voice.
- **Multi-Voice Retry Loop:** Seamless fallback on rate or tier limits.
- **Strict Design Rule:** Zero robotic synthetic voices (macOS say banned).

ELEVENLABS NEURAL VOICE PIPELINE

```
POST https://api.elevenlabs.io/v1/text-to-speech/{voice_id}
Headers: {'xi-api-key': KEY}, Payload: {'text': dialogue}
# Candidate shuffle: tries next approved voice on failure
# Generates duration-matched broadcast audio stream
```

PIPELINE SUMMARY: Touch Capture ⊐ Cloudflare QUIC Ingress ⊐ In-Memory WAV DSP (<4ms) ⊐ Local Gemma 3 Metal GPU (~1.3s) ⊐ ElevenLabs Voice (~1.2s) ⊐ In-Place Canvas Audio Playback **TOTAL LATENCY: ~2.5s (OPT.)**

Final outcome: The Cetacean Interface

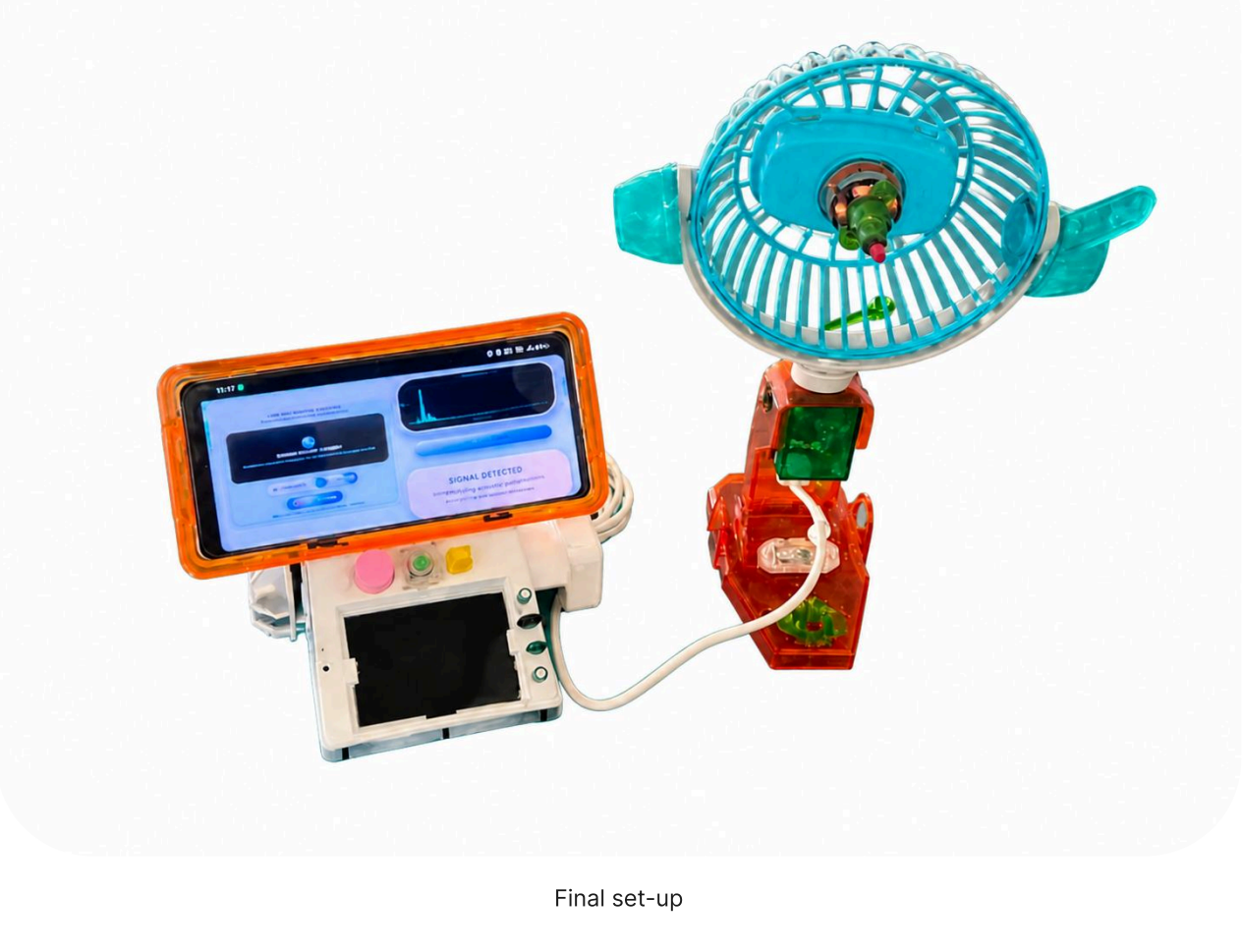

File change.
Rerun
Always rerun
CETACEAN INTERFACE
COCKPIT v2.4 • BIOACOUSTIC TRANSLATION
SYSTEM ONLINE
LOCAL AI
READY
LIVE OBSERVATION
ACOUSTIC SIGNAL & TELEMETRY
LIVE MULTIMODAL CAPTURE
LIVE TRANSLATION
UPLOAD RECORDING
ACOUSTIC SPECTROGRAM
REALTIME
SPECTRAL
AWAITING ACOUSTIC TRANSMISSION...
LIVE MULTIMODAL CAPTURE
SYNCHRONISED MICROPHONE & WEBCAM FEED
SENSORS OFFLINE • STANDBY
Microphone and camera disengaged. Click START CAPTURE to engage live feed.
DURATION: -- s
PEAK FREQ: -- Hz
AMPLITUDE: --
ANALYSE SIGNAL
FRONT CAMERA
BACK CAMERA
START CAPTURE
Awaiting sensor engagement.
STATE: IDLE
STANDBY • CAMERA & MICROPHONE OFF
INPUT: LIVE MULTIMODAL CAPTURE
HYDROPHONE: 16.0 kHz MONO
LOCAL LLM: GEMMA 3-4B
SIGNAL: STANDBY
COCKPIT
HYDROPHONE
TRANSLATION
SETTINGS
LOCAL MODEL: GEMMA 3-4B
CONNECTION: LOCAL
HYDROPHONE: REALTIME 16kHz MONO
ELEVENLABS SETTINGS

The digital interface

Artefact+Digital Interface

Cetacean Communication Station

The final prototype brings together the visual language established in the initial concept and the interactive functions developed through making. It allows a user to capture or upload a signal, examine acoustic information, provide visual context, and request an AI-generated interpretation. The artefact has evolved from an imagined physical communication station into a digital interface for exploring the possibility of interspecies communication.
Rather than presenting a solution to the problem of understanding cetaceans, the prototype stages an encounter with the limits of human interpretation. It asks not only how humans might communicate with another intelligence, but also what it means to listen when understanding is uncertain.



Final set-up